

LAB ASSIGNMENT - 01

Experiment Title :

Study of LAN Topology. Network Devices and Address Configuration.

Questions:-

1. Draw the Network Topology:-

- Identify and draw the LAN Topology of your organization
- Clearly label all components such as router, switch, Access points and end end devices (Pcs, laptops etc)

Solution :- The Academic Campus network uses a hybrid hierarchical topology that combines three distinct structural topologies.

It combines:

- Tree Topology
- star topology
- Ring topology

This network is further divided into three-layered network model which are

- core layer
- Distribution layer
- Access layer

i) Core layer:-

- Location:- server room (Room 211, 2nd floor)
- Equipments:-

i) Router and firewall:-

Manages external ISP connections, security filtering and external gateway routing.

ii) Layer-3 (st) switches:-

These switches are used in ring loop.

iii) Distribution layer:-

The main function is to interconnect the access floor layer and enforces local routing.

It Consists of 4 distribution switches:-

- i) R7 (Room 219, 2nd floor): Direct link to server room.
- ii) R1 (Room 801, 3rd floor): Aggregates upper floor traffic.
- iii) R2 (Room 201, 2nd floor): Interconnected with R1 and R7
- iv) R6 (Room 119, 1st floor): Aggregates lower floor and ground floor units.

Components used:-

- TP - Link TL-SG1218MPE
- TP - Link Omada SG3428 XMP

iii) Access layer:-

The Access layer directly connects end-user devices.

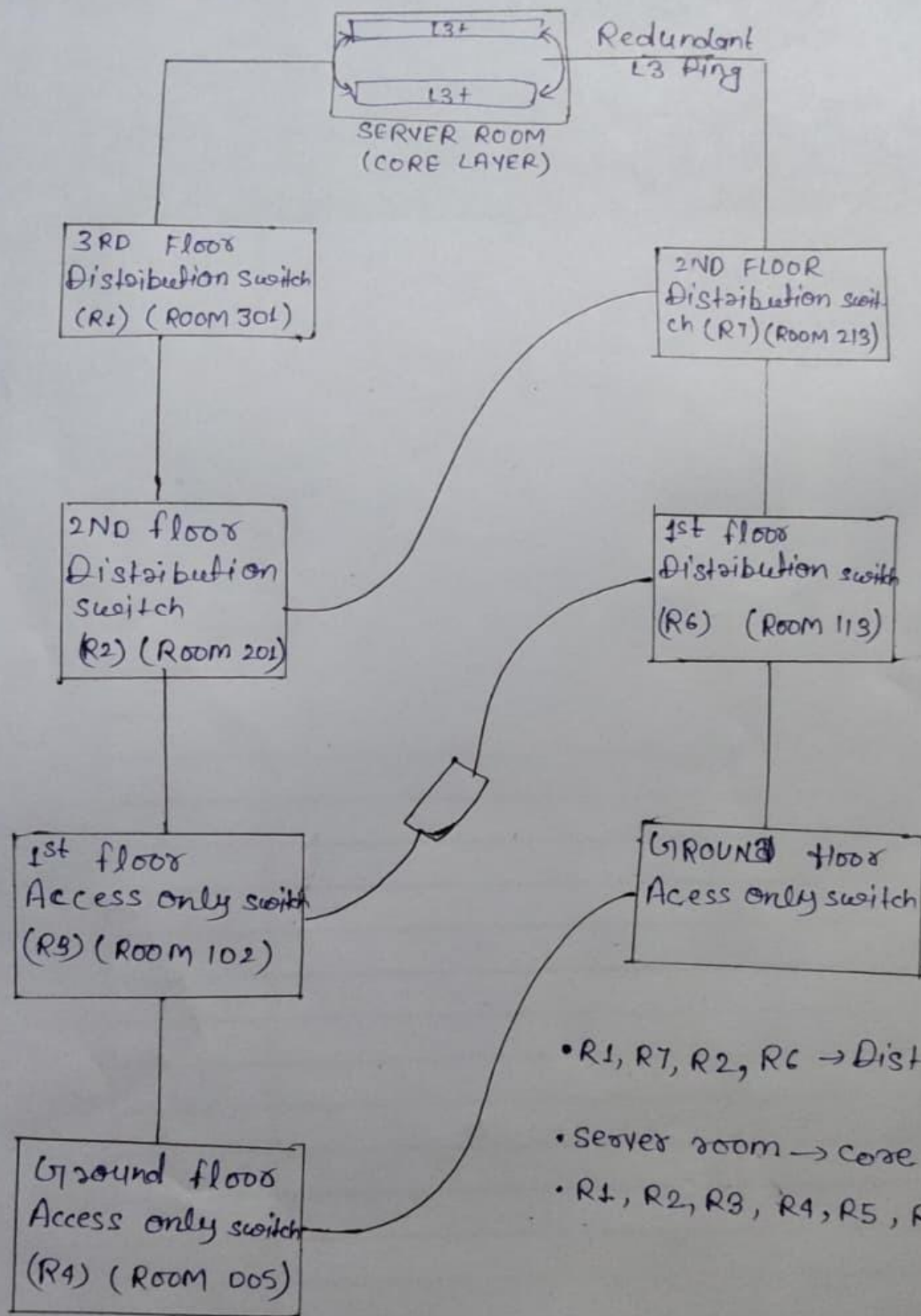
Connected Devices:-

- classroom Ethernet ports
- Laboratory Systems
- Faculty Room PCs.
- wireless Access points (APs)
- outdoor wireless APs

Access layer switches:-

- R1 → 3rd floor
- R2 → 2nd floor
- R3 → 1st floor
- R4 → Ground floor
- R5 → Ground floor
- R6 → 1st floor
- R7 → 2nd floor

ACADEMIC NETWORK TOPOLOGY



• R1, R7, R2, R6 → Distribution layer

• Server room → Core layer

• R1, R2, R3, R4, R5, R6, R7 → Access layer

Question - 02:- Identity Network Devices

- List all networking devices used in the LAN
- Mention the function of each device

Solution

1. Router and firewall:-

Location:- Central server room (Room 210, 2nd floor)

function

- (i) Acts as the primary gateway connecting the internal Campus LAN to the external Internet service providers
- (ii) It directs incoming and outgoing data traffic using IP routing.
- (iii) It filters data packets to protect the school network from unauthorized external threats.

2. Layer 3 Core switches:-

Location:- Distributed in central server room it is deployed as a dual-switch redundant pair.

function:- Distributed in central server room it is deployed as a dual-switch redundant pair

- (i) It forms the main high-speed backbone of the entire Campus network.
- (ii) Handles faster inter-VLAN routing (routing data between different floor, labs and departments).
- (iii) These switches are configured in ring topology loop to provide physical link redundancy.

3. TP-Link Omada S613428 XMP (Layered 2nd switches):

Location:- Distribution Layer nodes: R1, R2, R6, R7.

function (i) Aggregates traffic from the local access layer switches and user devices on each floor before uploading it to the Core layer.

- (ii) Provides managed Layer 2 capabilities like traffic prioritizing (QoS), security policies and network division.
- (iii) It supports inline PoE+ (Power over Ethernet) power directly to downline access points through its 24 gigabit ports.

4. TP-Link TL-SG1218MPE :-

Location :- Access Layer nodes : R3, R4, R5

Functions :-

- (i) Connects edge devices directly into the school network infrastructure.
- (ii) It offers 16 dedicated PoE+ ports specifically optimized to run network peripherals that require both data and power simultaneously.

5. wireless Access points (APs) :-

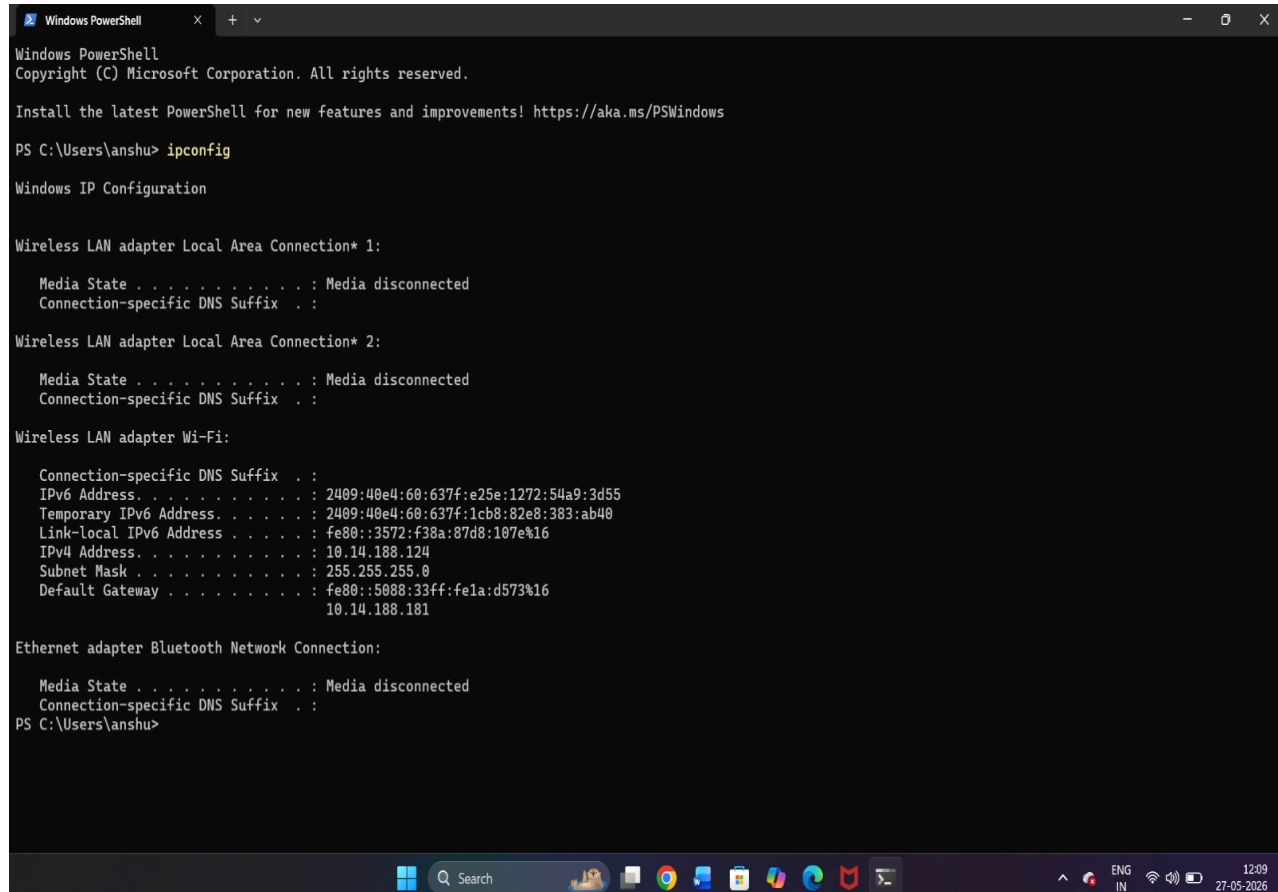
Location :- Distributed across all floors and outdoor zones.

Functions :-

- (i) It emits radio signals to provide wireless local area network (Wi-Fi) coverage across classrooms, labs etc.
- (ii) Bridges Mobile wireless client devices (laptops, smartphones, tablets) back into the physical wired ethernet distribution layer.

Question 3: Analyze Connection Types:

- Specify whether connections are:
 - Wired (Ethernet)
 - Wireless (Wi-Fi)
- Mention where each type is used.

A screenshot of a Windows PowerShell terminal window. The window title is "Windows PowerShell". The text inside shows the command "ipconfig" being executed. The output displays network configuration for two wireless adapters (Local Area Connection* 1 and 2) and a Wi-Fi adapter. The Wi-Fi adapter is connected, showing IPv4 and IPv6 addresses, subnet mask, and default gateway. The Ethernet adapter (Bluetooth Network Connection) is shown as disconnected. The taskbar at the bottom shows the Start button, search bar, and various application icons. The system tray on the right shows the date and time as 12:09 on 27-05-2026.

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Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\anshu> ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2409:40e4:60:637f:e25e:1272:54a9:3d55
    Temporary IPv6 Address. . . . . : 2409:40e4:60:637f:1cb8:82e8:383:ab40
    Link-local IPv6 Address . . . . . : fe80::3572:f38a:87d8:107e%16
    IPv4 Address. . . . . : 10.14.188.124
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::5088:33ff:fe1a:d573%16
                                10.14.188.181

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

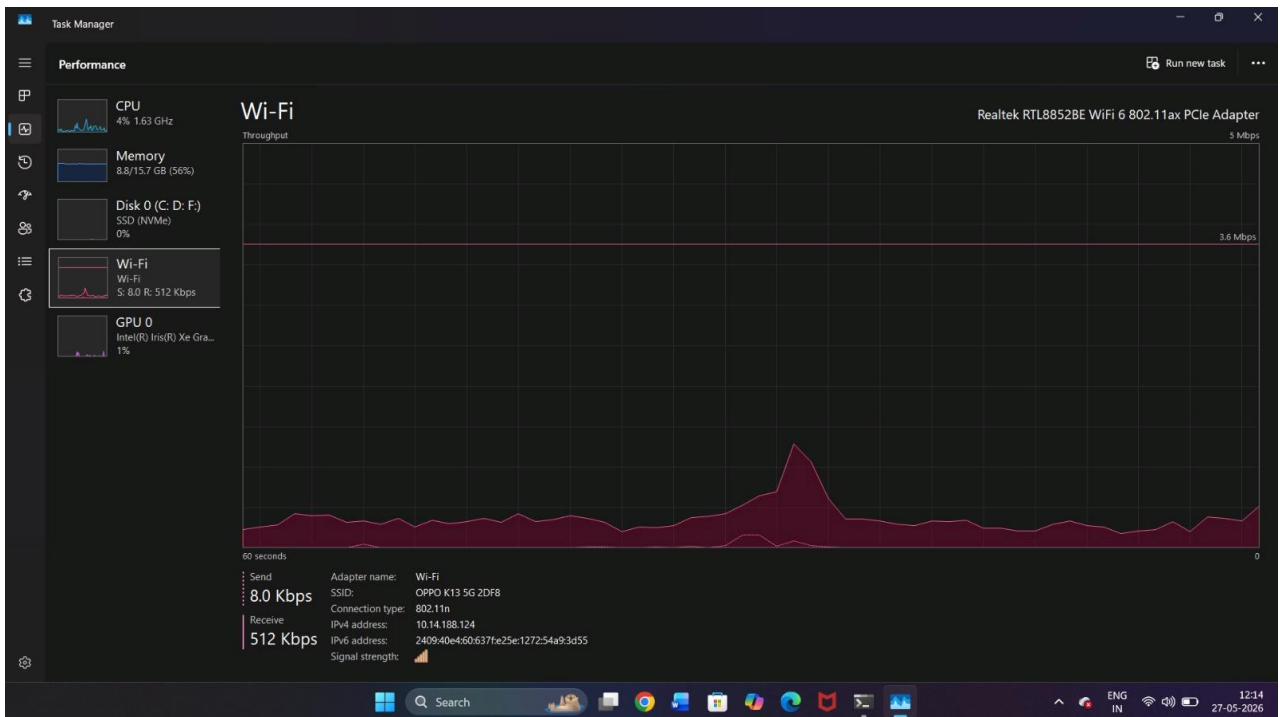
PS C:\Users\anshu>
```

⑨ By using ipconfig

QUESTION 4:-

Measure Network Speed:

- Find the link speed of:
 - Wired network (if available)
 - Wireless network
- Mention the tools/commands used (e.g., Task Manager, netsh; netsh wlan show interfaces or , etc.).



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Windows PowerShell
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PS C:\Users\anshu> netsh wlan show interfaces

There is 1 interface on the system:

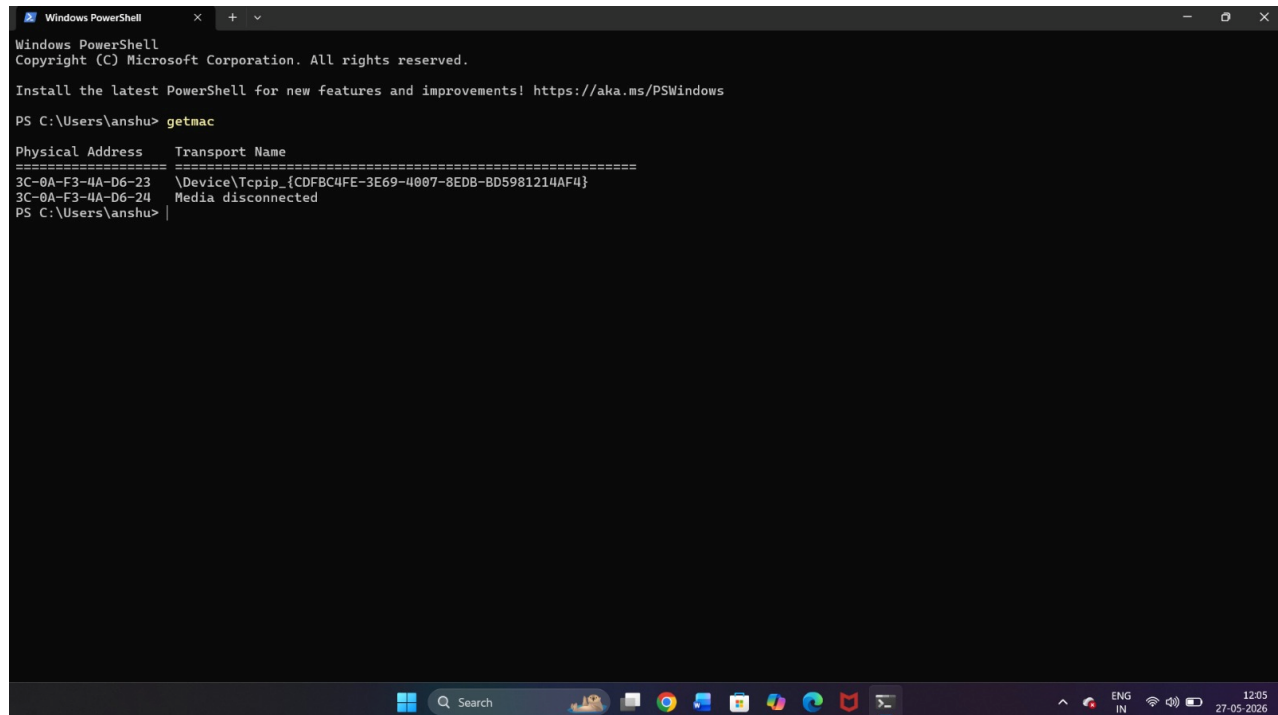
Name                : Wi-Fi
Description          : Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
GUID                 : cdfbc4fe-3e69-4007-8edb-bd5981214af4
Physical address     : 3c:0a:f3:4a:d6:23
Interface type       : Primary
State                : connected
SSID                : OPPO K13 5G 2DF8
AP BSSID             : 3e:bf:9c:ad:91:11
Band                 : 2.4 GHz
Channel              : 6
Connected Akm-cipher : [ akm = 00-0f-ac:00, cipher = 00-0f-ac:04 ]
Network type         : Infrastructure
Radio type           : 802.11n
Authentication       : WPA3-Personal (H2E)
Cipher               : CCMP
Connection mode      : Auto Connect
Receive rate (Mbps)  : 150
Transmit rate (Mbps) : 150
Signal               : 100%
Rssi                 : -29
Profile              : OPPO K13 5G 2DF8
QoS MSCS Configured  : 0
QoS Map Configured   : 0
QoS Map Allowed by Policy : 0

PS C:\Users\anshu>
```

->By using netsh wlan show interfaces

QUESTION 5. Find System Network Details:

- Determine and record the following of your computer:
 - MAC Address
 - IP Address (IPv4)
 - Subnet Mask
- Mention the command(s) used (e.g., ipconfig, getmac).



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Windows PowerShell
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PS C:\Users\anshu> getmac

Physical Address    Transport Name
=====
3C-0A-F3-4A-D6-23   \Device\NPF{C0FBC4FE-3E69-4007-8EDB-BD5981214AF4}
3C-0A-F3-4A-D6-24   Media disconnected
PS C:\Users\anshu> |
```

->By using getmac

Question-06 Explain observations:

Briefly explain:

Difference between wired & wireless speed

	wired speed	wireless speed
Data flow	wired supports full-duplex transmission	wireless on the other hand operates on half duplex
Signal Interference	Zero interference from Physical obstacles or other radio devices.	High interference from walls microwave ovens and neighbouring routers
Signal transmission	Electrical or light pulses inside a shielded cable	Radio frequencies (2.4 Gbps 5GHz-6GHz) through open air.
Latency	wired paths have minimal delay.	wireless path suffer from signal overhead and packet loss.

- Explain role of Subnet mask in networking.

Solution:— A Subnet mask is a 32-bit number used to manage network traffic routing. It identifies the network & host portion of an IP address.

• Separation:— It splits IP address into two parts

• Host ID:— It identifies the specific device address.

• Network ID:— It identifies the specific network address.

• Routing:— It tells router if a destination is local or is it is external.

• Efficiency:— It prevents unnecessary broadcast traffic across networks.